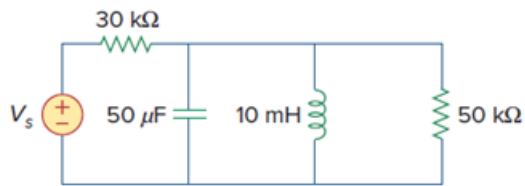


Problem

For the circuit shown in Fig. 14.84, find ω_0 , B , and Q , as seen by the voltage across the inductor.

Figure 14.84:



Step-by-step solution

Step 1 of 5

Consider the following circuit diagram:

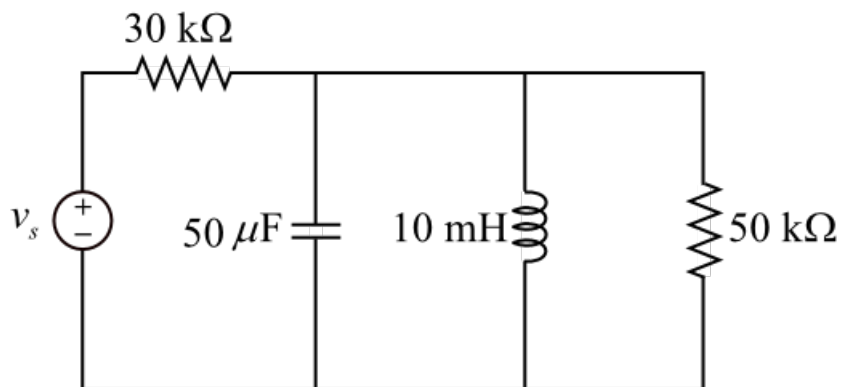


Figure 1

Step 2 of 5

A source transformation is the process of replacing a voltage source v_s in series with a resistor R by a current source i_s in parallel with a resistor R , or vice versa.

Apply source transformation to the circuit, we will get current source i_s in parallel with a resistor $30 \text{ k}\Omega$ as shown below.

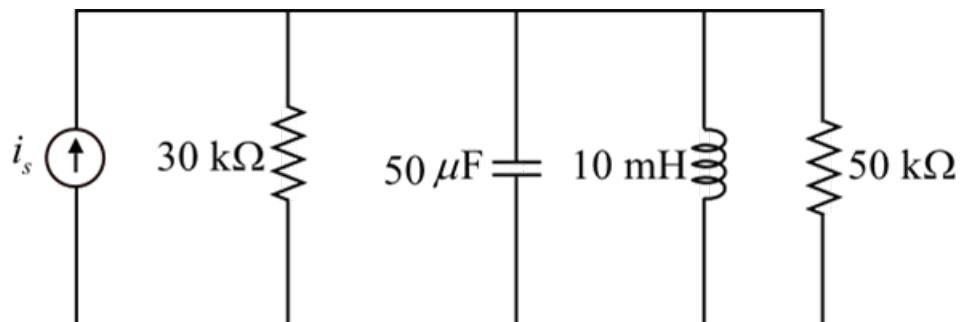


Figure 2

Step 3 of 5

To get equivalent resistance R , we combine resistors in parallel and in series. The $50\text{ k}\Omega$ and $30\text{ k}\Omega$ resistors are in parallel. So their equivalent resistance is:

$$\begin{aligned} R &= 30\text{ k}\Omega \parallel 50\text{ k}\Omega \\ &= \frac{30\text{ k} \times 50\text{ k}}{30\text{ k} + 50\text{ k}} \\ &= 18.75\text{ k}\Omega \end{aligned}$$

Redraw, figure 2 by replacing resistors $50\text{ k}\Omega$ and $30\text{ k}\Omega$ with equivalent resistance $18.75\text{ k}\Omega$.

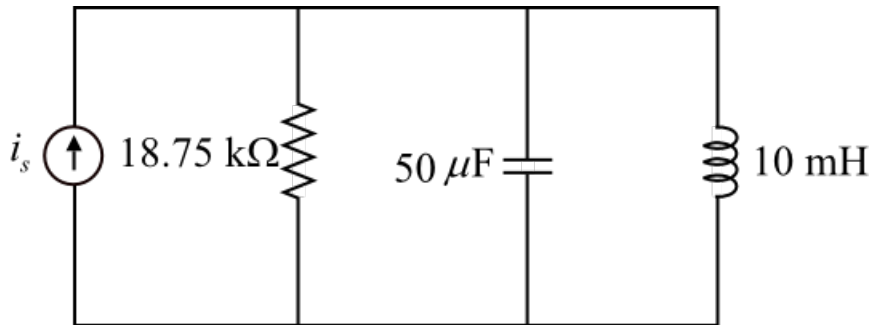


Figure 3

Step 4 of 5

Figure 3 is a parallel RLC circuit.

Write the expression for resonant frequency ω_0 .

$$\begin{aligned} \omega_0 &= \frac{1}{\sqrt{LC}} \\ &= \frac{1}{\sqrt{(10 \times 10^{-3})(50 \times 10^{-6})}} \\ &= \frac{1}{\sqrt{0.5 \times 10^{-6}}} \\ &= \frac{1}{7.07 \times 10^{-4}} \\ &= 1414.21\text{ rad/s} \\ &= 1.4142\text{ krad/s} \end{aligned}$$

Thus, resonant frequency ω_0 is 1.414 krad/s.

Step 5 of 5

Write the expression for quality factor of a parallel resonance circuit:

$$Q = \omega_0 RC$$

Substitute $18.75 \text{ k}\Omega$ for R , 50μ for C and 1414.2 for ω_0 in the expression $\omega_0 RC$ to get the quality factor.

$$\begin{aligned} Q &= (1414.2)(18.75 \times 10^3)(50 \times 10^{-6}) \\ &= 1325.82 \end{aligned}$$

Therefore the quality factor Q is $\boxed{1325.82}$.

Write the expression for bandwidth of the circuit.

$$B = \frac{\omega_0}{Q}$$

Substitute 1325.82 for Q and 1414.21 for ω_0 values in the above expression.

$$\begin{aligned} B &= \frac{1414.21}{1325.82} \\ &= 1.067 \text{ rad/s} \end{aligned}$$

Therefore the bandwidth of the circuit is $\boxed{1.067 \text{ rad/s}}$.